

HR / POWER

Report di identificazione

Run 507a006c6c474f518fd9447673cbb487

Dataset: G2_Ambigui

Creazione: 2026-09-28T18:35:25.220308+00:00 | Stato: completed

Struttura: p1d_full | $\tau dx/dt + x = K (P(t-L)-P_0)$; $HR_{hat} = B+x$

Inizializzazione: estimated_equilibrium | $P_0 = \text{mean}(P_{pre})$; $HR_0 = HR(0)$; $x(0) = HR_0 - B$; B estimated; $P(t < 0) = P_0$

Dataset: 8 segmenti; train richiesti: 7; risultati: 7; validi: 7; falliti: 0.

Ambito: fit in-sample sui soli segmenti train. Val/test non utilizzati. Aggregati non pesati per durata, calcolati solo sui record fitted; n indica i valori numerici disponibili. n.d. = non disponibile o non applicabile. Nessuna conclusione fisiologica automatica.

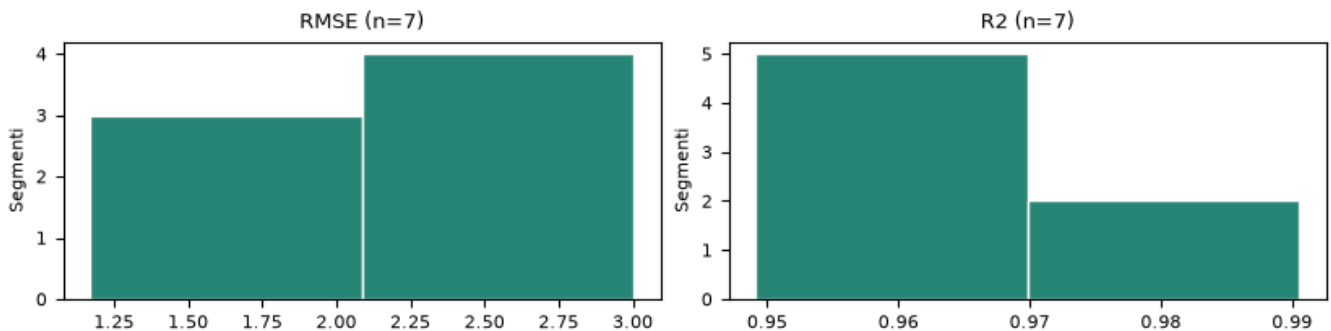
Pre-window richiesta: si; durata: 10 s. P0: pre_window_mean; HR0: first_segment_sample

Pre-window disponibili: 7 / 7 segmenti train inventariati.

Qualita della traiettoria

RMSE, MAE e bias in bpm; R2 e amplitude_ratio adimensionali. Bias positivo: HR osservata maggiore della prevista.

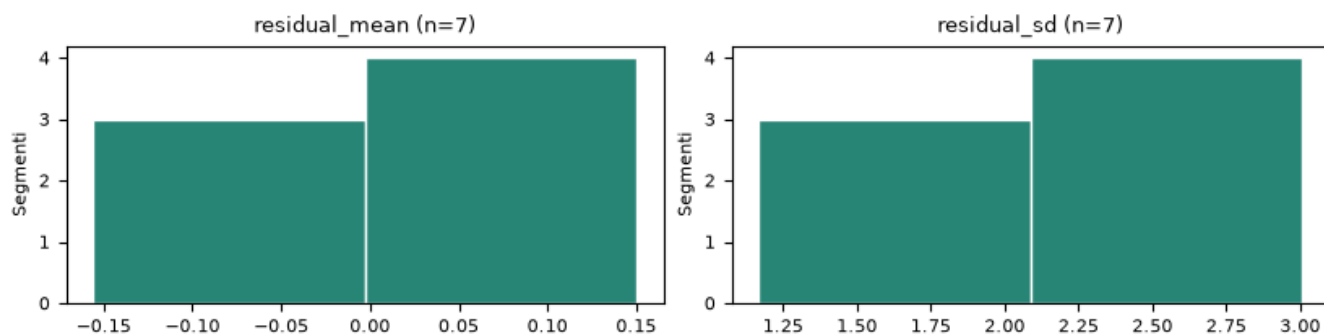
Metrica	n	Media	Mediana	SD	Min	Max
RMSE	7	2.0063	2.1368	0.65161	1.1696	3.0027
MAE	7	1.5586	1.6449	0.59424	0.84954	2.5767
bias	7	-0.022522	0.0037045	0.12092	-0.15593	0.15019
R2	7	0.965	0.95942	0.015785	0.94912	0.99053
amplitude_ratio	7	0.96785	0.96926	0.037183	0.90313	1.0141



Residui

e = HR osservata - HR prevista. SD campionaria (ddof=1); ACF centrata, lag in campioni. Le segnalazioni sono conservate e non impongono nuove esclusioni.

Metrica	n	Media	Mediana	SD	Min	Max
residual_mean	7	-0.022522	0.0037045	0.12092	-0.15593	0.15019
residual_sd	7	2.0077	2.137	0.65377	1.1671	3.0064
residual_autocorrelation_lag1	7	0.92597	0.94	0.042384	0.86082	0.97294
residual_autocorrelation_lag5	7	0.50469	0.47558	0.19233	0.26982	0.75627



Segnalazione	Segmenti
autocorrelated_residuals	7
multistart_disagreement	7

Multistart

Ricerca: profile_multistart. I tentativi comprendono gli eventuali raffinamenti da griglia; n_multistart indica solo gli start originali.

Start registrati: 112; convergenti: 112. Accordo di tutti gli start: 0 / 7 fit validi.

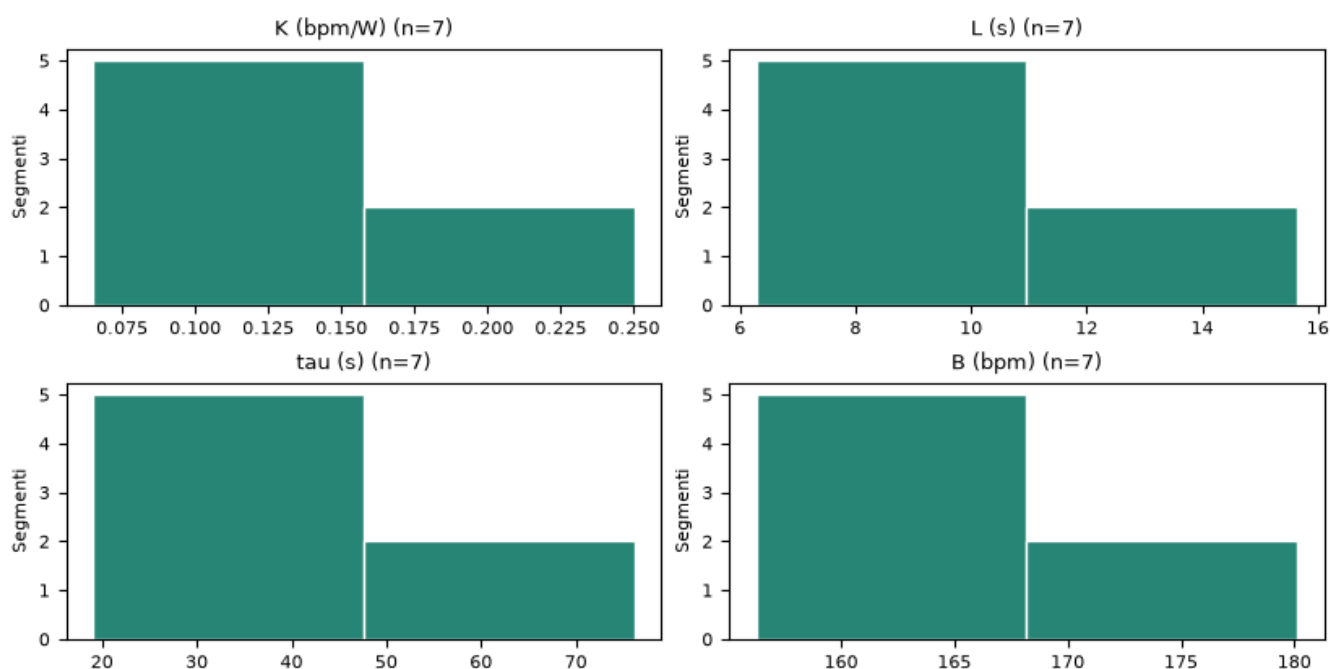
Diagnostica	n	Media	Mediana	SD	Min	Max
SSE	112	1806.1	1178.8	1744.5	247.6	9824.1
nfev	112	15.911	11	24.714	4	192

Distribuzione e precisione dei parametri

SE e CI95 locali di Wald, condizionati ai riferimenti iniziali. RSE in percentuale. Bounds, autocorrelazione e rango ridotto possono limitare la precisione. Un parametro fissato non ha SE/CI stimabili.

Parametro	n	Media	Mediana	SD	Min	Max
K (bpm/W)	7	0.13014	0.099003	0.073058	0.065058	0.25007
L (s)	7	9.4953	8.241	3.3778	6.2782	15.641
tau (s)	7	38.653	30.452	20.203	19.049	76.153
B (bpm)	7	165.33	162.5	7.9574	156.25	180.13

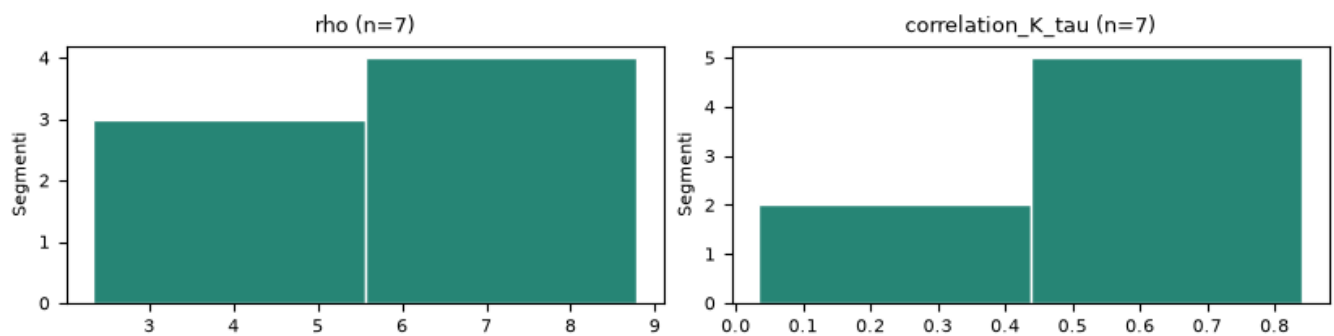
Parametro	SE mediana	RSE mediana %	AI bound	Vicino al bound
K	0.0054221	6.1861	0	0
L	0.25691	3.171	0	0
tau	0.89065	2.9438	0	0
B	0.60823	0.37428	0	0



Identificabilita

- $\rho = (T-L)/\tau$: tempo disponibile dopo il ritardo in unita di τ . Valori piccoli descrivono una frazione breve del transitorio; nessuna soglia automatica.
- Correlazioni parametriche vicine a -1 o +1 segnalano compensazione locale tra stime; non misurano la qualita della traiettoria.
- Rango Jacobiano: confrontare con p , il numero di parametri liberi. Rango $< p$ indica direzioni non identificabili localmente.
- Multistart: SSE simili con parametri differenti indicano ambiguita. CI e correlazioni restano approssimazioni locali.

Diagnostica	n	Media	Mediana	SD	Min	Max
ρ	7	5.7409	5.6971	2.1358	2.3342	8.7829
correlation_K_tau	7	0.51838	0.56185	0.31182	0.034074	0.84084
Jacobian rank	7	4	4	0	4	4
Jacobian condition	7	1490	1139.4	1277.4	300.73	3306.4



Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G2/260428165622__laps_007-007.csv

ID: 078e8a10a675cbc122240510 | Stato: fitted

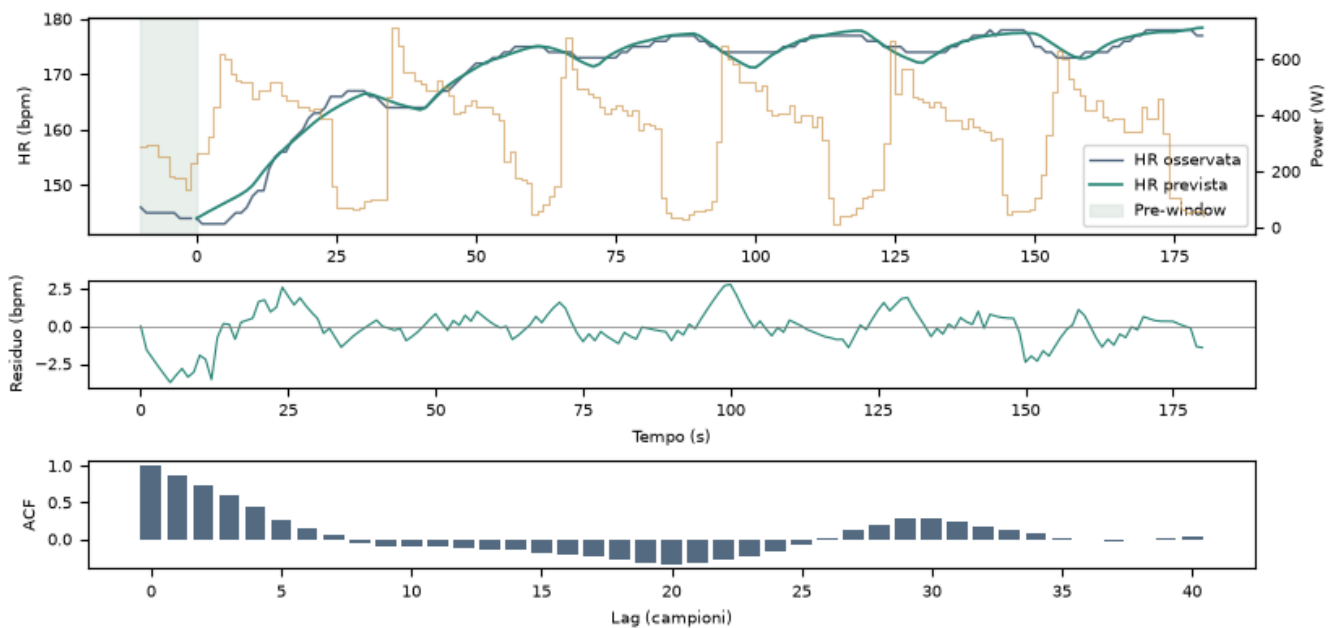
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=230 W; HR0=144 bpm.

HR(0) osservata fissata; B=166.84 bpm stimato; x(0)=-22.843 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.1696	0.84954	-0.11584	0.98233	0.98334

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.11584	1.1671	0.86082	0.26982



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.099003	0.0061244	0.086999 / 0.11101	6.1861	no	no
L	6.2782	0.19908	5.888 / 6.6684	3.171	no	no
tau	39.008	0.95418	37.138 / 40.879	2.4461	no	no
B	166.84	0.57878	165.71 / 167.98	0.3469	no	no

rho=4.4534; Jacobian rank=4/4; cond=1139.4; accordo multistart=no.

Correlazioni: {"K_L": -0.0965779041915741, "K_tau": 0.7851961179032373, "K_B": -0.9653595467801516, "L_tau": -0.19451916789052298, "L_B": 0.07227215879964424, "tau_B": -0.6440715258587302}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 247.6 / 911.23

Griglia: 42441 coppie L/tau; SSE multistart originale: 247.6; selezione: multistart.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G2/250729173622__laps_020-025.csv

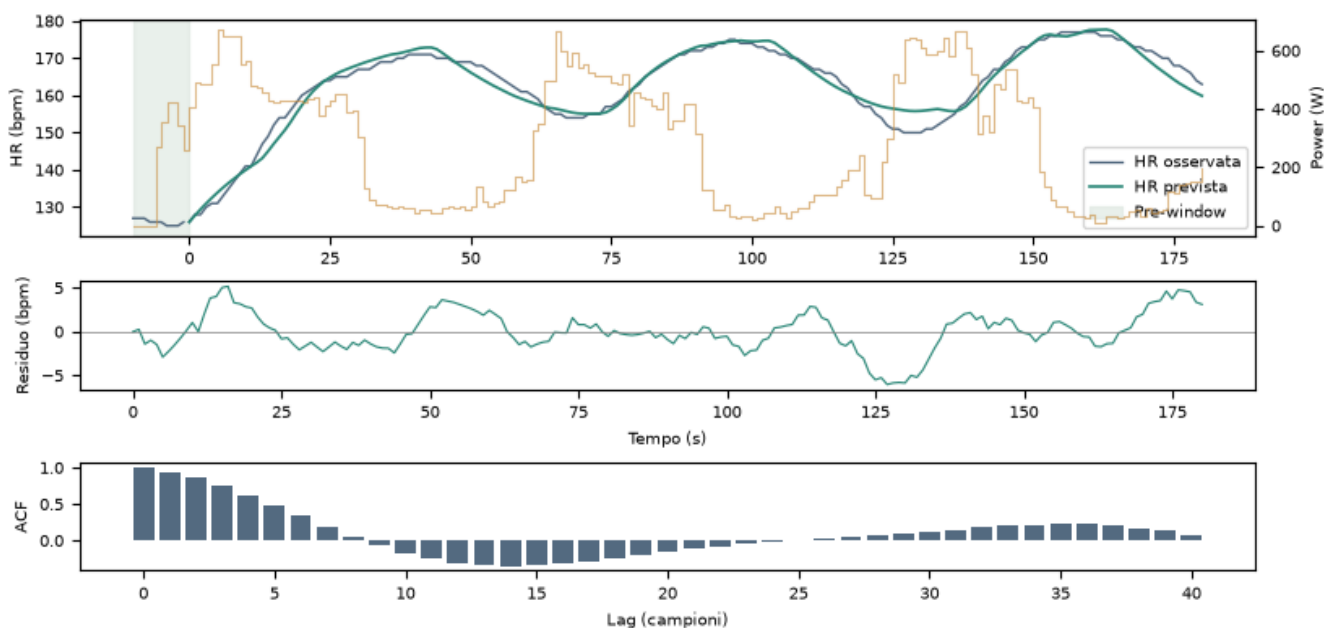
ID: 0b778497f663f33e9336d0d2 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=207.8 W; HR0=126 bpm.

HR(0) osservata fissata; B=160.19 bpm stimato; x(0)=-34.193 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.2373	1.7093	0.020166	0.95942	1.0141
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.020166	2.2434	0.94	0.47558	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.083444	0.0035507	0.076484 / 0.090403	4.2552	no	no
L	12.695	0.25691	12.192 / 13.199	2.0236	no	no
tau	19.049	0.72281	17.632 / 20.466	3.7945	no	no
B	160.19	0.26888	159.67 / 160.72	0.16785	no	no

rho=8.7829; Jacobian rank=4/4; cond=526.69; accordo multistart=no.

Correlazioni: {"K_L": -0.5216200127235209, "K_tau": 0.8408366559124727, "K_B": -0.6607401213889057, "L_tau": -0.6277445315853731, "L_B": 0.21480546243206008, "tau_B": -0.3872777286335407}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 905.97 / 9824.1

Griglia: 42441 coppie L/tau; SSE multistart originale: 905.97; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G2/260904062108__laps_005-005.csv

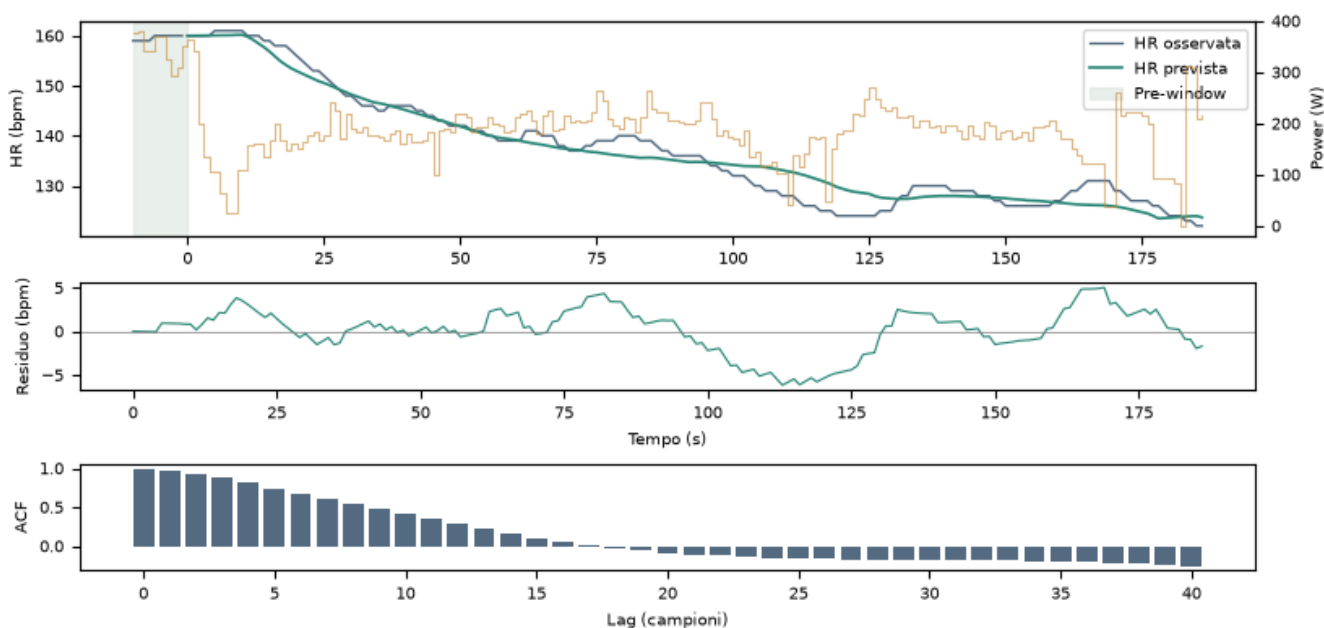
ID: 3fef587e2dca92c4c6ea2960 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=346 W; HR0=160 bpm.

HR(0) osservata fissata; B=161.12 bpm stimato; x(0)=-1.1248 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.5107	1.9115	0.15019	0.94912	0.94078
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.15019	2.513	0.9675	0.74193	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.25007	0.036397	0.17873 / 0.32141	14.555	no	no
L	8.241	0.94309	6.3925 / 10.089	11.444	no	no
tau	76.153	4.4976	67.337 / 84.968	5.906	no	no
B	161.12	5.7842	149.79 / 172.46	3.5899	no	no

rho=2.3342; Jacobian rank=4/4; cond=3306.4; accordo multistart=no.

Correlazioni: {"K_L": -0.4354113962806347, "K_tau": 0.034073599501529175, "K_B": 0.9871084895495453, "L_tau": -0.4353294872309965, "L_B": -0.38953663262410276, "tau_B": -0.11637299782435748}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 1178.8 / 1462.5

Griglia: 42441 coppie L/tau; SSE multistart originale: 1178.8; selezione: multistart.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G2/260428165622__laps_002-002.csv

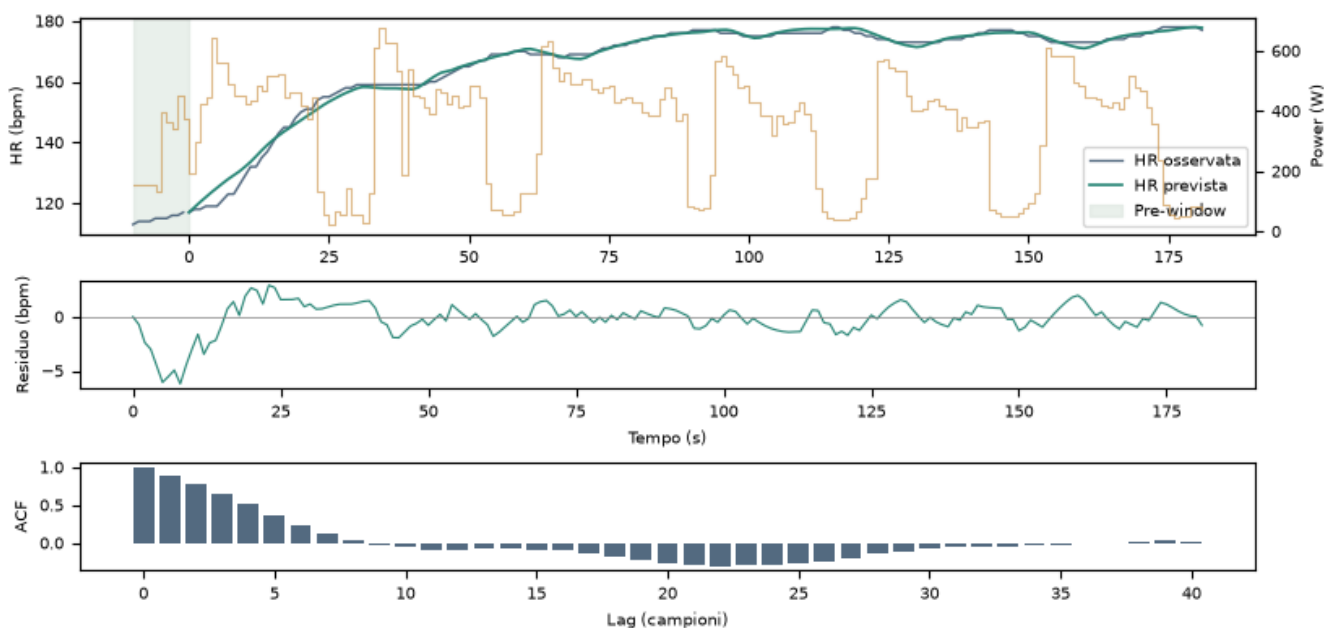
ID: 5fe6dc3820d603507896d5c6 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=268.6 W; HR0=117 bpm.

HR(0) osservata fissata; B=170.24 bpm stimato; x(0)=-53.242 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.45	1.0065	-0.14991	0.99053	0.99927
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.14991	1.4462	0.88978	0.35651	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.075209	0.00446	0.066467 / 0.08395	5.9302	no	no
L	7.5113	0.25382	7.0138 / 8.0088	3.3792	no	no
tau	30.452	0.40074	29.667 / 31.237	1.316	no	no
B	170.24	0.36532	169.53 / 170.96	0.21459	no	no

rho=5.6971; Jacobian rank=4/4; cond=300.73; accordo multistart=no.

Correlazioni: {"K_L": -0.05245013029432715, "K_tau": 0.4532027713165411, "K_B": -0.8876849737859459, "L_tau": -0.14049989163547064, "L_B": 0.026806042911933028, "tau_B": -0.13025791055926234}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 382.67 / 1096.2

Griglia: 42441 coppie L/tau; SSE multistart originale: 382.67; selezione: multistart.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G2/260428165622__laps_009-009.csv

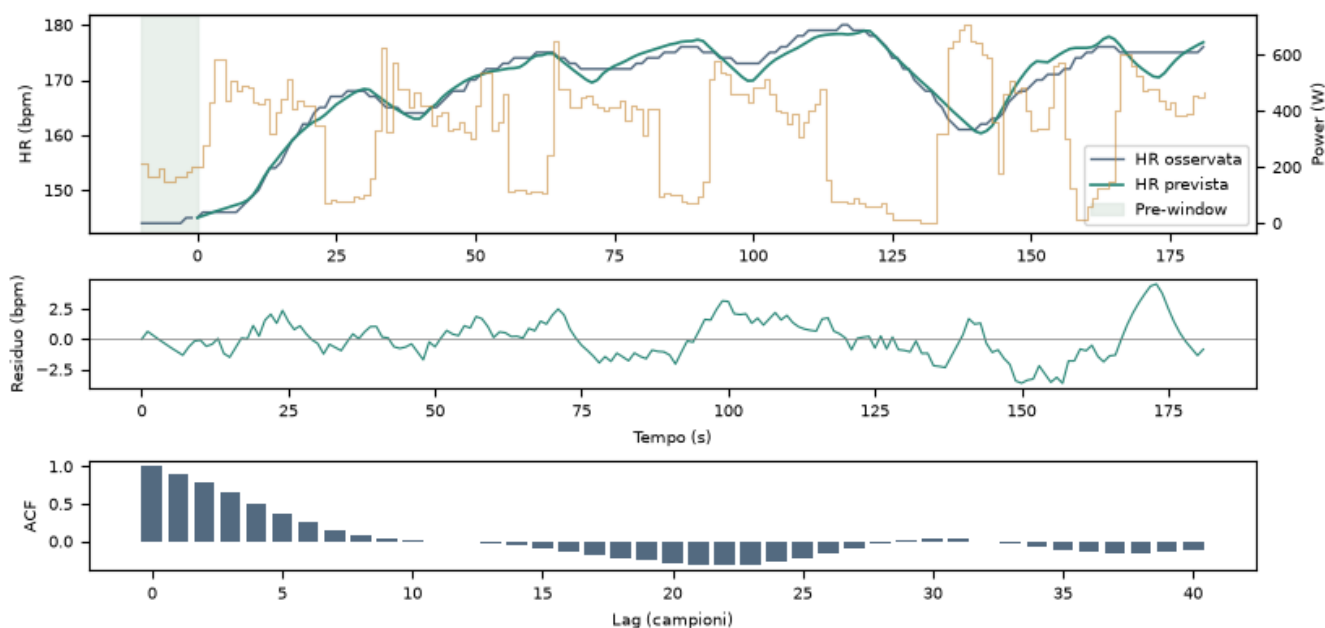
ID: 6ea1418c3be2b82de49a9a5e | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=175.8 W; HR0=145 bpm.

HR(0) osservata fissata; B=156.25 bpm stimato; x(0)=-11.245 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.5369	1.2121	0.0037045	0.96473	0.96926
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.0037045	1.5411	0.90198	0.36277	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.12328	0.0054221	0.11265 / 0.13391	4.3982	no	no
L	7.6006	0.15867	7.2896 / 7.9116	2.0876	no	no
tau	30.255	0.89065	28.509 / 32	2.9438	no	no
B	156.25	0.71254	154.85 / 157.64	0.45604	no	no

rho=5.7313; Jacobian rank=4/4; cond=1191.4; accordo multistart=no.

Correlazioni: {"K_L": -0.2572286175685109, "K_tau": 0.7631320396862458, "K_B": -0.9752168175776479, "L_tau": -0.33165135997415424, "L_B": 0.24336631340279455, "tau_B": -0.6685272552239216}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 429.89 / 3109.4

Griglia: 42441 coppie L/tau; SSE multistart originale: 429.89; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G2/260428165622__laps_004-004.csv

ID: ce9ef96a872b65e8f4aca862 | Stato: fitted

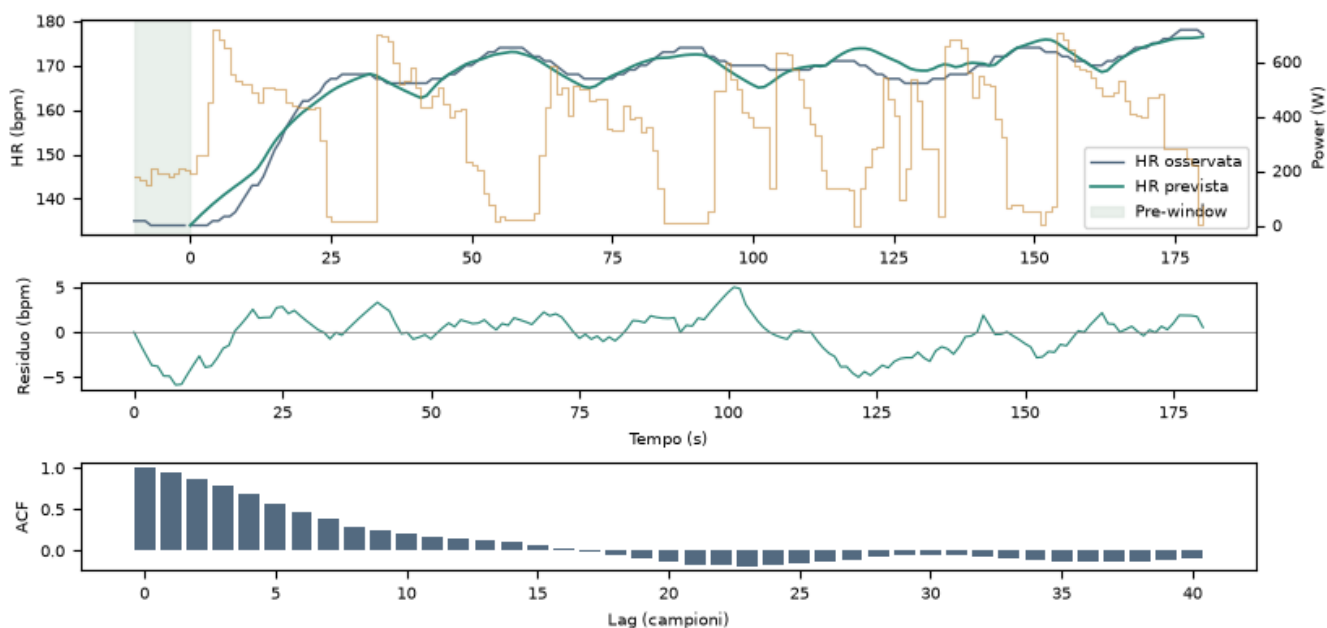
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=188 W; HR0=134 bpm.

HR(0) osservata fissata; B=162.5 bpm stimato; x(0)=-28.504 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.1368	1.6449	-0.15593	0.95018	0.96512

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.15593	2.137	0.94879	0.56997



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.065058	0.004492	0.056253 / 0.073862	6.9046	no	no
L	8.4992	0.26293	7.9839 / 9.0146	3.0936	no	no
tau	21.717	0.70772	20.33 / 23.104	3.2589	no	no
B	162.5	0.60823	161.31 / 163.7	0.37428	no	no

rho=7.8972; Jacobian rank=4/4; cond=669.16; accordo multistart=no.

Correlazioni: {"K_L": -0.09046364578572379, "K_tau": 0.561849312276771, "K_B": -0.9371692885831931, "L_tau": -0.2108982287687832, "L_B": 0.0744521049204671, "tau_B": -0.3702915596219514}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 826.41 / 1935.8

Griglia: 42441 coppie L/tau; SSE multistart originale: 826.41; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::Ambigui_G2/260716153012__laps_009-009.csv

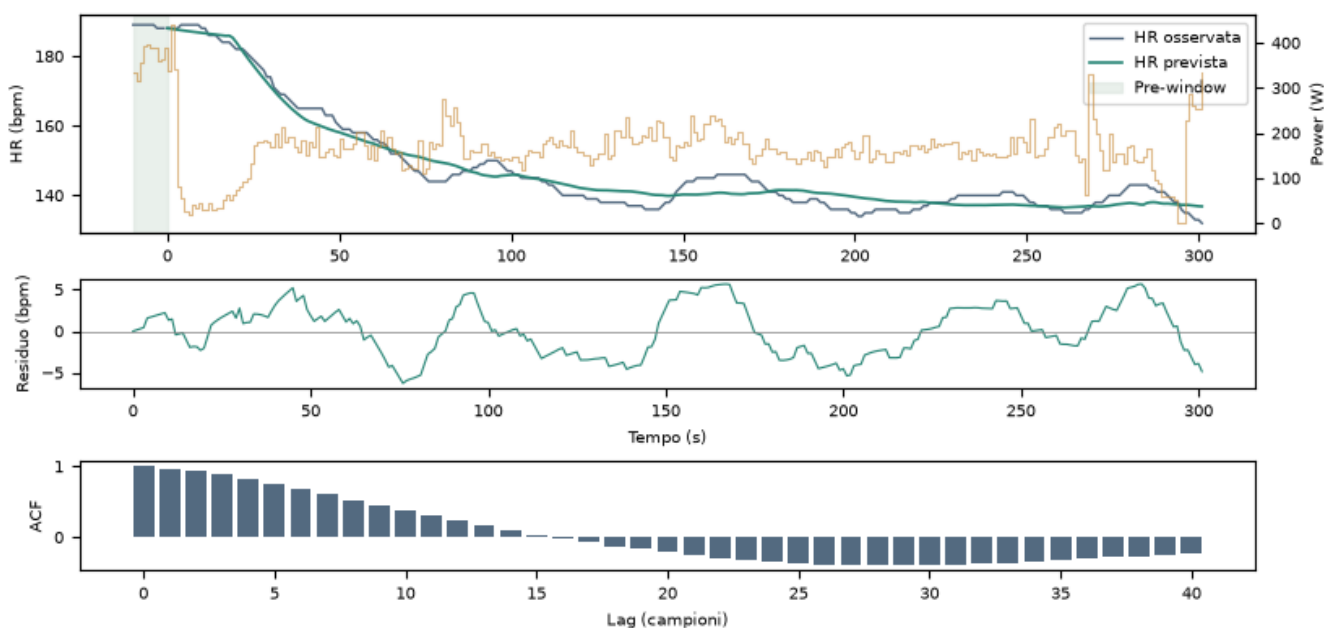
ID: e74d4a8c1e199b7882060430 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=368.9 W; HR0=188 bpm.

HR(0) osservata fissata; B=180.13 bpm stimato; $x(0)=7.8744$ bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
3.0027	2.5767	0.089965	0.95871	0.90313
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.089965	3.0064	0.97294	0.75627	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.21494	0.016944	0.18173 / 0.24815	7.8832	no	no
L	15.641	0.79961	14.074 / 17.209	5.1121	no	no
tau	53.94	1.4573	51.084 / 56.796	2.7017	no	no
B	180.13	3.3551	173.55 / 186.7	1.8627	no	no

rho=5.2903; Jacobian rank=4/4; cond=3296.4; accordo multistart=no.

Correlazioni: {"K_L": -0.5597618586852645, "K_tau": 0.19033517966049435, "K_B": 0.9959931187542698, "L_tau": -0.5277415130209212, "L_B": -0.5443007552370915, "tau_B": 0.12788156033642128}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 2723 / 4488.3

Griglia: 42441 coppie L/tau; SSE multistart originale: 2723; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Configurazione completa dell'esperimento

Configurazione, split, sorgenti, hash e ambiente della run. I CSV originali restano conservati nel blob store. Manifest integrale e risultati sono inclusi nell'export tabellare ZIP.

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{
  "schema_version": 3,
  "id": "507a006c6c474f518fd9447673cbb487",
  "manifest_filename": "20260928T183525Z_G2_Ambigui_p1d_raw_70-10-20_s42_n8-f42_507a006c.json",
  "created_at": "2026-09-28T18:35:25.220308+00:00",
  "status": "completed",
  "split": {
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    "unit": "activity",
    "seed": 42,
    "percentages": [
      70.0,
      10.0,
      20.0
    ],
    "algorithm": "sorted groups; numpy PCG64 shuffle; largest remainder on group counts v1",
    "assignments": {
      "train": [
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        "5fe6dc3820d603507896d5c6",
        "ce9ef96a872b65e8f4aca862",
        "078e8a10a675cbc122240510",
        "6ea1418c3be2b82de49a9a5e",
        "e74d4a8c1e199b7882060430",
        "0b778497f663f33e9336d0d2"
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      "test": [
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      1
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    "actual_percentages": [
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      0.0,
      12.5
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    "warnings": [
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  },
  "signal": "raw",
  "columns": [
    "power_w",
    "heart_rate_bpm"
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  "model_structure": "p1d_full",
  "pre_window_seconds": 10,
  "use_pre_window": true,
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      "hr_finite": true,
      "first_time": -10.0,
      "last_time": -1.0
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      "available": true,
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      "samples": 10,
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      "first_time": -10.0,
      "last_time": -1.0
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      "samples": 10,
      "signals_finite": true,
      "power_finite": true,
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      "first_time": -10.0,
      "last_time": -1.0
    }
  ],
}
```

```

{
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  "seconds": 10,
  "samples": 10,
  "signals_finite": true,
  "power_finite": true,
  "hr_finite": true,
  "first_time": -10.0,
  "last_time": -1.0
},
{
  "segment_id": "6ea1418c3be2b82de49a9a5e",
  "available": true,
  "seconds": 10,
  "samples": 10,
  "signals_finite": true,
  "power_finite": true,
  "hr_finite": true,
  "first_time": -10.0,
  "last_time": -1.0
},
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  "available": true,
  "seconds": 10,
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"B_method": "estimated",
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      {
        "key": "tau",
        "unit": "s"
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      {
        "key": "B",
        "unit": "bpm"
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    ],
    "equation": "tau dx/dt + x = K (P(t-L)-P0); HRhat = B+x",
    "source": "PID richiesto nel protocollo utente; nessuna replica bibliografica rivendicata",
    "initial_conditions": "P0=mean(Ppre); HR0=HR(0); x(0)=HR0-B; B estimated; P(t<0)=P0",
    "input_reconstruction": "zero-order hold, integrazione esatta; ritardo continuo in secondi",
    "uncertainty": "SSE/(N-p) inv(J.T J); p=parametri liberi; CI95 Wald non troncati; condizionati a P0 e HR0",
    "delay_domain": "0 <= L <= min(Lmax,T); T=t[-1]-t[0]; Lmax=null significa T; Lmax=0 fissa L=0",
    "identification_validity": "Convergenza e compatibilit\u00e0 numerica/temporale; non certifica precisione o identificabilit\u00e0",
    "residual_conventions": "e=HR-HRhat; SD ddof=1; centered ACF denominator=sum(e_centered**2); lag in samples",
    "multistart_conventions": "PCG64; first start nominal; log-uniform K/tau or gamma, uniform L/B; optional deterministic profile grid; select minimum SSE across all refinements",
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    "rho_convention": "(T-L)/tau for PID full; null for short_transient; no exclusion thresholds",
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